CHAPTER 1

INTRODUCTION

The increasing demand for electricity and the depletion of conventional energy resources have created a significant need for sustainable and renewable power generation technologies. Rapid industrialization and the widespread use of electronic devices have considerably increased global energy consumption. Conventional energy sources such as coal, petroleum, and natural gas are limited in availability and contribute to environmental pollution. Therefore, there is a growing need for alternative energy solutions that are economical, environmentally friendly, and independent of conventional power grids.

Human-powered energy generation has emerged as an effective approach for producing electricity through mechanical effort. Pedal-powered systems convert the rotational motion generated by human pedaling into electrical energy using an electromechanical generator. These systems are simple, cost-effective, and particularly suitable for locations where access to electricity is limited. In addition, they promote physical activity while simultaneously generating useful electrical power for small-scale applications.

The Pedal Powered Electricity Generator developed in this project utilizes a pedal mechanism coupled with a 100 RPM DC geared motor operating as a generator. The electrical output produced by the motor is regulated using an LM2596 DC-DC buck converter to ensure a stable voltage supply. The regulated output is connected to a USB charging module, enabling the charging of mobile phones, LED lamps, rechargeable batteries, and other low-voltage electronic devices.

The proposed system is portable, economical, and environmentally sustainable. It can serve as an emergency power source during power outages and can be effectively utilized in rural areas, disaster relief camps, educational institutions,

fitness centers, and outdoor applications. By utilizing human effort as a renewable energy source, the project demonstrates a practical and sustainable solution for electricity generation while promoting awareness of clean energy technologies.

1.1 Introduction to Human Powered Energy Systems

Human-powered energy systems are technologies that utilize mechanical energy produced by human effort to perform useful work or generate electrical power. Historically, human energy has been used for transportation, agricultural activities, water pumping, and operating mechanical equipment. With advancements in electromechanical systems, human effort can now be effectively harnessed for electricity generation using generators, dynamos, and DC motors.

Pedal-powered systems represent one of the most efficient forms of human-powered energy generation due to their ability to utilize the stronger muscles of the human legs. The average adult can comfortably produce between 50 W and 150 W of mechanical power for short durations, making pedal mechanisms suitable for small-scale electricity generation applications.

Human-powered electricity generation systems consist of four major elements:

- Mechanical Input Unit (Pedals and Crank)
- Power Transmission System (Chain and Sprocket)
- Electrical Generator Unit
- Power Conditioning and Output Unit

These systems are particularly beneficial in regions lacking reliable electricity infrastructure. They provide a sustainable means of generating electricity without fuel consumption and contribute to environmental conservation by reducing dependence on fossil fuels.

1.2 Need for Sustainable Electricity Generation

Electricity has become an essential requirement for modern society. The operation of communication systems, healthcare equipment, educational tools, and household appliances depends heavily on a reliable power supply. However, many rural and remote regions continue to experience inadequate access to electricity, while urban areas frequently encounter power interruptions and increasing energy costs.

Conventional electricity generation methods present several challenges:

- Depletion of fossil fuel reserves.
- Environmental pollution.
- Greenhouse gas emissions.
- Increasing electricity demand.
- High operational costs.
- Dependence on centralized power grids.

Renewable energy technologies such as solar, wind, and hydroelectric power have provided alternative solutions; however, these methods often depend on environmental conditions. Human-powered energy generation offers a unique advantage by producing electricity whenever mechanical effort is available.

1.3 Problems with Conventional Portable Power Sources

Problems Associated with Conventional Power Sources:

1. Dependence on external charging infrastructure.
2. Limited operational life of batteries.
3. Environmental hazards due to battery disposal.
4. Fuel consumption in portable generators.
5. Noise pollution.

6. High maintenance requirements.
7. Increased operational costs.
8. Carbon emissions.
9. Limited availability during emergencies.
10. Reduced reliability in remote locations.

Diesel generators, although effective, contribute significantly to air pollution and require continuous fuel supply. Similarly, rechargeable batteries eventually degrade over time and require replacement. Power banks depend entirely on prior charging and cannot provide unlimited energy.

The Pedal Powered Electricity Generator eliminates many of these limitations by generating electricity directly from human effort. Since the system does not rely on fossil fuels or external charging sources, it offers an environmentally friendly and economically viable alternative for portable power generation.

1.4 Importance of Renewable Energy

Renewable energy plays a crucial role in addressing global energy challenges and mitigating the adverse effects of climate change. The transition from conventional energy sources to renewable technologies is essential for achieving long-term sustainability and ensuring energy security for future generations.

- Reduction in greenhouse gas emissions.
- Conservation of natural resources.
- Improved energy security.
- Sustainable economic development.
- Reduced environmental impact.
- Enhanced public health.
- Increased accessibility to electricity.

Human-powered electricity generation represents a unique category of renewable energy because it directly utilizes human mechanical effort without requiring external natural resources. It serves as an excellent example of sustainable engineering practices and promotes awareness regarding energy conservation.

1.5 Objectives of the Project

The primary objective of the project is to design and develop a Pedal Powered Electricity Generator capable of converting human mechanical energy into regulated electrical energy for charging low-voltage electronic devices.

1. To design a portable pedal-powered electricity generation system.
2. To convert human mechanical energy into electrical energy using a 100 RPM DC geared motor.
3. To regulate the generated voltage using an LM2596 DC-DC buck converter.
4. To provide a stable USB output for charging electronic devices.
5. To evaluate the performance of the system under different pedaling conditions.
6. To analyze voltage output with respect to pedaling speed.
7. To develop a cost-effective and environmentally friendly power generation system.
8. To reduce dependence on conventional electricity sources.
9. To promote awareness regarding renewable and sustainable energy technologies.
10. To demonstrate the practical application of energy conversion principles.

1.6 Methodology

1. **Problem Identification:**
    - Identify the need for portable and sustainable electricity generation.
2. **Literature Review:**
    - Study existing human-powered energy generation systems.
    - Analyze the performance of DC motors used as generators.
3. **System Design:**
    - Design the pedal mechanism and transmission arrangement.
    - Select suitable electrical and mechanical components.
4. **Fabrication:**
    - Assemble the pedal frame, chain drive, and generator setup.
    - Install the buck converter and USB charging module.
5. **Circuit Integration:**
    - Connect the DC geared motor, LM2596 module, and output terminals.
    - Ensure proper voltage regulation and electrical safety.
6. **Testing:**
    - Measure output voltage under various pedaling speeds.
    - Evaluate charging capability for different devices.
7. **Performance Analysis:**
    - Analyze system efficiency.
    - Record voltage and current characteristics.
    - Compare experimental results with expected performance.
8. **Documentation:**
    - Prepare technical drawings, performance graphs, and final project documentation.

CHAPTER 2

LITERATURE SURVEY

The growing demand for renewable and sustainable energy sources has encouraged researchers to explore alternative methods of electricity generation. Human-powered energy generation systems have gained considerable attention due to their simplicity, portability, and environmental benefits. Pedal-powered electricity generation, in particular, has emerged as an effective means of converting mechanical energy into electrical energy using readily available components such as DC motors, generators, and power conditioning circuits.

Several studies have investigated the use of bicycle mechanisms and pedal-driven systems for small-scale electricity generation. These systems have demonstrated the capability to produce sufficient electrical power for charging portable electronic devices, emergency lighting systems, and battery storage units. Researchers have also focused on improving energy conversion efficiency, voltage regulation, and user comfort to enhance the practicality of pedal-powered generators.

Recent developments in power electronics have significantly improved the performance of human-powered electricity generation systems. Voltage regulators, DC-DC converters, and USB charging modules have enabled stable and reliable power delivery to electronic devices. Additionally, advancements in lightweight materials and compact mechanical designs have increased the portability and usability of such systems in rural and emergency applications.

The literature survey presented in this chapter reviews previous studies related to human-powered energy systems, pedal-driven mechanisms, DC motors operating as generators, energy harvesting technologies, and portable charging applications. The review helps identify existing limitations, technological

advancements, and the motivation behind developing a Pedal Powered Electricity Generator for sustainable electricity production.

2.1 Human Powered Energy Generation

Human-powered energy generation refers to the process of converting mechanical energy produced by human effort into useful electrical energy. Researchers have demonstrated that the average individual can continuously generate between 50 W and 100 W of mechanical power through pedaling activities. This capability makes pedal-powered systems suitable for low-power applications such as mobile charging, LED lighting, and battery charging.

Several human-powered systems have been developed using bicycle mechanisms coupled with DC generators. These systems are widely recognized for their low operating costs, ease of maintenance, and environmental sustainability. Studies have shown that human-powered generators can provide an effective alternative to conventional energy sources in remote and off-grid regions.

Furthermore, human-powered systems promote energy awareness and encourage sustainable practices among users. Their ability to generate electricity without fuel consumption makes them particularly attractive for educational demonstrations and renewable energy initiatives.

2.2 Pedal Driven Mechanisms

Pedal-driven mechanisms play a vital role in the efficient transmission of mechanical energy. The pedal assembly typically consists of a crankshaft, chain, sprockets, and bearings that convert human effort into rotational motion. Researchers have investigated various transmission systems to maximize energy transfer efficiency and minimize mechanical losses.

Studies indicate that chain-driven mechanisms exhibit higher efficiency compared to belt-driven arrangements due to reduced slippage and improved torque transmission. The use of appropriate gear ratios has also been found to significantly influence the speed and output characteristics of the generator.

Modern pedal-powered systems incorporate lightweight materials and ergonomic designs to improve user comfort during prolonged operation. Experimental investigations have demonstrated that optimized pedal mechanisms can enhance overall system performance while reducing user fatigue.

2.3 DC Motor as Generator

DC motors are commonly used as generators in small-scale electricity generation systems due to their simplicity, affordability, and availability. When a DC motor shaft is rotated mechanically, electromagnetic induction causes an electrical voltage to be generated across its terminals. This principle has been extensively studied and applied in numerous renewable energy projects.

Researchers have reported that permanent magnet DC motors exhibit stable output characteristics and are well suited for pedal-powered applications. The generated voltage is directly proportional to the rotational speed of the motor, making pedaling speed a critical factor influencing system performance.

Several studies have analyzed the efficiency of DC motors operating under generator conditions. Results indicate that appropriate voltage regulation and load matching are necessary to ensure stable operation and maximize power output. Consequently, DC motors continue to be widely utilized in educational and prototype renewable energy systems.

2.4 Energy Harvesting Techniques

Energy harvesting is the process of capturing and converting energy available from various sources into usable electrical power. Common energy harvesting techniques include solar energy, wind energy, thermal energy, vibration energy, and human-powered energy generation. These technologies play a significant role in reducing dependence on conventional energy resources and promoting sustainable development. Among these methods, human-powered energy harvesting has gained considerable attention due to its ability to generate electricity independent of environmental conditions.

Human-powered energy harvesting systems utilize mechanical energy produced by human activities such as walking, cycling, and hand cranking to generate electrical power. Unlike solar and wind energy systems, pedal-powered generators can operate whenever human effort is available, making them highly reliable for emergency and portable applications. Recent advancements in power electronics, energy storage devices, and voltage regulation techniques have further enhanced the efficiency and practicality of energy harvesting systems, enabling their use in mobile charging, LED lighting, and other low-power electronic applications.

2.5 Research Gap and Motivation

Although numerous studies have been conducted on pedal-powered electricity generation systems, several limitations still exist in currently available designs. Most existing systems are bulky, costly, and primarily intended for demonstration purposes rather than practical day-to-day applications. In addition, many of these systems lack proper voltage regulation, making them unsuitable for charging sensitive electronic devices such as mobile phones and USB-powered equipment. The absence of compact designs and efficient power management circuits reduces their usability in remote and emergency situations.

Furthermore, limited research has focused on developing portable and user-friendly pedal-powered generators that can provide a stable output voltage with minimal pedaling effort. Existing systems often require high rotational speeds to generate usable power, leading to user fatigue and reduced operational efficiency. Therefore, this project aims to bridge these gaps by developing a low-cost, portable, and efficient Pedal Powered Electricity Generator integrated with a 100 RPM DC geared motor, LM2596 buck converter, and USB charging module to deliver a reliable and sustainable power solution.

CHAPTER 3
PRINCIPLES OF PEDAL POWERED ELECTRICITY GENERATION

The Pedal Powered Electricity Generator operates on the principle of converting human mechanical energy into electrical energy through electromagnetic induction. Human effort applied to the pedals creates rotational motion, which is transmitted to a DC geared motor using a chain and sprocket mechanism. The DC motor acts as a generator and produces electrical energy based on its rotational speed.

The generated electrical power is regulated using an LM2596 buck converter to provide a stable output suitable for charging mobile phones, LED lamps, and other low-voltage devices. The system demonstrates the practical application of energy conversion, power transmission, and voltage regulation while providing an environmentally friendly and sustainable solution for electricity generation.

3.1 Introduction

Electricity generation through human effort has become an important area of research in renewable energy technologies. Pedal-powered systems are among the most efficient human-powered mechanisms due to their ability to utilize the strong muscles of the legs to generate continuous rotational motion. Such systems are particularly useful in remote locations and emergency situations where access to grid electricity is limited.

The Pedal Powered Electricity Generator converts mechanical energy produced by pedaling into electrical energy using a DC geared motor. The generated voltage is subsequently regulated to provide a stable output suitable for various low-power applications. The simplicity, affordability, and portability of the system make it a practical solution for sustainable electricity generation.

3.2 Mechanical Energy Conversion

Mechanical energy conversion is the process of transforming one form of mechanical energy into another usable form. In the proposed system, human muscular energy is converted into rotational mechanical energy through the pedal mechanism.

1. Human effort is applied to the pedals.
2. The crank mechanism converts linear force into rotational motion.
3. The chain and sprocket transmit rotational energy to the DC motor shaft.
4. The rotating shaft drives the motor, causing electrical energy generation.

3.3 Electromagnetic Induction Principle

The operation of the Pedal Powered Electricity Generator is based on the principle of electromagnetic induction. According to Faraday's Law, an electromotive force (EMF) is induced when there is relative motion between a conductor and a magnetic field. In this project, the pedal mechanism rotates the shaft of the 100 RPM DC geared motor, causing its armature windings to move within the magnetic field and generate electrical energy.

The generated voltage depends on factors such as the rotational speed of the motor, magnetic field strength, and electrical load connected to the system. Higher pedaling speeds result in higher voltage generation. The generated output is then regulated using an LM2596 DC-DC buck converter to provide a stable voltage suitable for charging low-voltage electronic devices.

3.4 Power Generation Characteristics

The power generation characteristics of the Pedal Powered Electricity Generator are influenced by various operating parameters. Experimental studies indicate

that the output voltage increases approximately linearly with pedaling speed under no-load conditions.

Pedaling Speed (RPM)	Generated Voltage (V)
40	8
60	14
80	20
100	26
120	30

The generated electrical power can be utilized directly or stored in rechargeable batteries for future use. The inclusion of an LM2596 buck converter enables stable voltage regulation, making the system suitable for powering USB devices and LED lighting applications.

3.5 Importance of Human Powered Systems

Human-powered systems offer several advantages over conventional electricity generation methods. They operate without fossil fuels, produce zero emissions, and require minimal maintenance. Furthermore, they encourage sustainable practices and increase public awareness regarding renewable energy technologies.

- Environmentally friendly operation.
- Low manufacturing cost.
- Grid-independent power generation.
- Portability and ease of transportation.
- Reduced carbon footprint.
- Educational and research applications.

Pedal-powered generators are particularly useful in rural communities and disaster-affected regions where conventional electricity infrastructure may be unavailable. Their ability to provide emergency power makes them an attractive solution for sustainable energy applications.

3.6 Sustainable Energy Approach

Sustainable energy refers to energy systems that fulfill present energy requirements without affecting future generations. Human-powered electricity generation supports sustainability by utilizing renewable human effort as an energy source.

The Pedal Powered Electricity Generator reduces dependence on fossil fuels, promotes renewable energy utilization, and encourages energy conservation. By integrating mechanical engineering concepts with renewable energy technologies, the proposed system provides an affordable, environmentally friendly, and sustainable solution for electricity generation.

CHAPTER 4
SYSTEM DESIGN AND COMPONENT DESCRIPTION

The Pedal Powered Electricity Generator is designed to convert human mechanical energy into electrical energy through an integrated mechanical and electrical system. The system consists of a pedal assembly, chain and sprocket mechanism, DC geared motor, voltage regulation unit, and USB charging module. The proper selection and integration of these components ensure efficient power generation and stable output for charging low-voltage electronic devices.

The overall design emphasizes portability, simplicity, and cost-effectiveness. Mechanical energy produced by pedaling is transmitted to the generator through a chain drive mechanism, where it is converted into electrical energy. The generated voltage is regulated using a buck converter before being supplied to the output devices. The proposed design minimizes energy losses while ensuring reliable operation under various loading conditions.

This chapter presents the overall system architecture, component specifications, block diagram description, and design considerations adopted for the development of the Pedal Powered Electricity Generator.

4.1 Introduction

System design plays a crucial role in determining the performance and reliability of any engineering project. The Pedal Powered Electricity Generator has been designed to provide a compact and efficient electricity generation solution using readily available mechanical and electrical components.

The primary objective of the design is to maximize energy conversion efficiency while maintaining user comfort and operational safety. The integration of voltage

regulation and USB charging modules further enhances the practical usability of the system.

4.2 Overall System Architecture

- Human Power Input System
- Mechanical Transmission System
- Electricity Generation Unit
- Voltage Regulation Unit
- Output Charging Module

4.3 Block Diagram Description

The block diagram of the Pedal Powered Electricity Generator illustrates the sequence of energy conversion from human effort to electrical output. Initially, the user applies mechanical force to the pedal assembly, which produces rotational motion. This motion is transmitted through the chain and sprocket mechanism to the 100 RPM DC geared motor.

The DC geared motor operates as a generator and converts the mechanical energy into electrical energy through electromagnetic induction. The generated voltage is then regulated using an LM2596 DC-DC buck converter to provide a stable output. Finally, the regulated power is supplied to the USB charging module, which can be used to charge mobile phones, LED lamps, and other low-voltage electronic devices.

4.4 List of Components

S. No	Component Name	Quantity
1	Bicycle Pedal Assembly	1
2	100 RPM DC Geared Motor	1
3	LM2596 Buck Converter	1
4	USB Charging Module	1
5	Connecting Wires	As Required
6	Nuts and Bolts	As Required
7	Mild Steel Frame	1

4.5 Component Description

4.5.1 Bicycle Pedal Assembly

The pedal assembly serves as the primary input mechanism of the system. It converts human muscular effort into rotational motion and provides continuous power input to the transmission system. The use of bicycle pedals ensures user comfort and efficient energy transfer.



FIGURE 4.1

4.5.2 100 RPM DC Geared Motor

The 100 RPM DC geared motor functions as the electricity generation unit. When mechanically rotated, it produces electrical energy based on the principle of electromagnetic induction.

Specifications:

Parameter	Value
Rated Speed	100 RPM
Operating Voltage	12–30 V
Type	DC Geared Motor
Shaft Type	Metal Shaft
Application	Power Generation

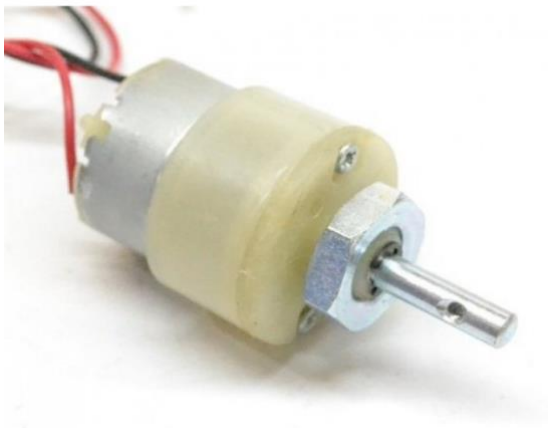


FIGURE 4.2

4.5.3 LM2596 Buck Converter

The LM2596 DC-DC buck converter is used to regulate the voltage generated by the DC geared motor and provide a stable output for charging electronic devices. It accepts an input voltage ranging from **4 V to 40 V** and converts it into a constant, adjustable output voltage according to the application requirement. The

module is capable of delivering a maximum output current of **3 A**, making it suitable for powering low-voltage devices such as mobile phones, LED lamps, and rechargeable batteries. With a conversion efficiency of up to **92%**, the LM2596 minimizes power loss and improves the overall performance of the Pedal Powered Electricity Generator by ensuring a reliable and efficient power supply.



FIGURE 4.3

4.5.5 USB Charging Module

The USB charging module is used to provide a standard 5 V USB output for charging various electronic devices. It converts the regulated voltage from the buck converter into a safe output suitable for USB-powered equipment. The module is compatible with mobile phones, power banks, LED lamps, rechargeable batteries, and other portable devices. It ensures stable and reliable charging while protecting connected devices from voltage fluctuations. The USB charging module enhances the practicality and user convenience of the Pedal Powered Electricity Generator.



FIGURE 4.4

4.6 Design Considerations

The design of the Pedal Powered Electricity Generator was developed by considering several important factors to ensure efficient and reliable performance. The system was designed to achieve efficient power transmission with minimum mechanical losses while providing a stable voltage output for charging electronic devices. User comfort during pedaling, low manufacturing cost, and ease of maintenance were also considered to make the system practical and economical. In addition, the design emphasizes portability, compactness, electrical safety, and structural stability to ensure reliable operation under different working conditions.

4.7 System Advantages

- Eco-friendly operation.
- Portable and lightweight design.
- Low initial and maintenance cost.
- No fuel requirement.
- Grid-independent operation.

- Suitable for emergency applications.
- Easy fabrication and assembly.
- Promotes renewable energy awareness.

The proposed system successfully combines mechanical engineering principles with renewable energy technologies to provide an efficient and sustainable electricity generation solution. The design methodology adopted ensures reliable performance while maintaining simplicity and cost-effectiveness.

CHAPTER 5

CIRCUIT DESIGN AND CONNECTIONS

The circuit design of the Pedal Powered Electricity Generator is an important part of the system, as it ensures the efficient conversion and regulation of electrical energy generated from human pedaling. The electrical circuit consists of a 100 RPM DC geared motor, an LM2596 DC-DC buck converter, a USB charging module, connecting wires, and an LED indicator. These components are interconnected to provide a stable and safe electrical output suitable for charging low-voltage electronic devices.

The DC geared motor acts as a generator and produces variable DC voltage depending on the pedaling speed. Since the generated voltage is not constant, it is regulated using the LM2596 buck converter, which provides a stable output voltage. The regulated power is then supplied to the USB charging module, enabling the charging of mobile phones, LED lamps, rechargeable batteries, and other USB-powered devices. This chapter describes the circuit diagram, electrical connections, voltage regulation, and safety measures adopted in the proposed system.

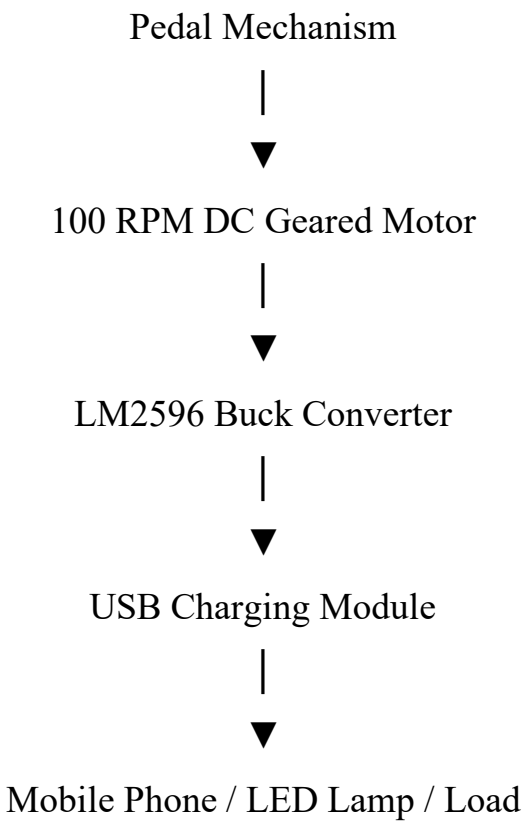
5.1 Introduction

The electrical circuit is designed to convert the variable voltage generated by the DC motor into a stable and usable power supply. Proper electrical connections ensure efficient energy transfer and protect the connected devices from voltage fluctuations. The use of a buck converter and USB charging module improves the reliability and safety of the system.

5.2 Circuit Diagram

The circuit consists of a DC geared motor connected to an LM2596 buck converter. The output of the buck converter is connected to a USB charging module, which provides a standard 5 V output for charging electronic devices.

Figure 5.2 Circuit Diagram



5.3 Electrical Connections

The electrical connections are made using insulated copper wires to minimize power losses and ensure safe operation. The positive and negative terminals of the DC geared motor are connected to the input terminals of the LM2596 buck converter. The output terminals of the buck converter are then connected to the USB charging module. Finally, the USB charging module provides a regulated 5 V output for charging various electronic devices. Proper polarity is maintained throughout the circuit to prevent damage to the components.

5.4 DC Geared Motor Wiring

The 100 RPM DC geared motor functions as the electricity generation unit. During pedaling, the motor shaft rotates and generates DC voltage through electromagnetic induction. The motor terminals are directly connected to the input of the LM2596 buck converter. Secure wiring and proper insulation are provided to ensure safe and efficient transmission of electrical energy.

5.5 LM2596 Buck Converter Configuration

The LM2596 buck converter is configured to regulate the variable voltage generated by the DC motor. It accepts an input voltage ranging from 4 V to 40 V and converts it into a stable output voltage. The output voltage is adjusted to 5 V using the onboard potentiometer to meet the requirements of USB-powered devices. The high conversion efficiency of the converter minimizes energy loss and improves the overall performance of the system.

5.6 USB Charging Circuit

The USB charging module receives the regulated 5 V output from the buck converter and provides a standard USB interface for charging electronic devices. It ensures safe charging by maintaining a constant output voltage and current. The module is compatible with mobile phones, rechargeable batteries, LED lamps, and other USB-powered devices, making the system practical for everyday use.

5.7 Power Supply Arrangement

The power supply arrangement begins with the mechanical energy produced by pedaling. The DC geared motor converts this mechanical energy into electrical energy. The generated voltage is regulated using the LM2596 buck converter to provide a stable output. The regulated power is then supplied to the USB charging module for charging low-voltage electronic devices safely and efficiently.

CHAPTER 6
WORKING PRINCIPLE

The Pedal Powered Electricity Generator operates by converting human mechanical energy into electrical energy through the principle of electromagnetic induction. When the user pedals the system, rotational motion is generated and transmitted to the DC geared motor through a chain and sprocket mechanism. The DC motor functions as a generator and produces electrical energy proportional to the speed of rotation.

The generated voltage is variable in nature and depends on the pedaling speed. Therefore, an LM2596 DC-DC buck converter is used to regulate the voltage and provide a stable output suitable for charging electronic devices. The regulated output is supplied to a USB charging module, enabling the safe charging of mobile phones, LED lamps, and rechargeable batteries.

This chapter explains the operational sequence and the various stages involved in the conversion of mechanical energy into usable electrical energy.

6.1 Introduction

The working principle of the Pedal Powered Electricity Generator is based on the efficient utilization of human effort for electricity generation. The system combines mechanical and electrical components to produce a reliable and sustainable source of energy. The use of a DC geared motor, voltage regulator, and USB charging module ensures stable operation and improved system performance.

6.2 Pedaling Mechanism

The pedaling mechanism serves as the primary source of mechanical energy in the system. When the user applies force to the pedals, rotational motion is produced. This motion is transmitted through the chain and sprocket arrangement to the shaft of the DC geared motor.

The efficiency of the system depends on the consistency of the pedaling speed and the effectiveness of the power transmission mechanism.

6.3 Power Generation Process

The power generation process begins when the DC geared motor shaft rotates due to the applied mechanical energy. The motor operates as a generator and converts the rotational energy into electrical energy through electromagnetic induction. The generated voltage increases with an increase in pedaling speed and can reach approximately 30 V under suitable operating conditions.

6.4 Voltage Regulation Process

The electrical output obtained from the DC geared motor is not constant and varies according to the speed of pedaling. To ensure safe operation of electronic devices, the generated voltage is supplied to an LM2596 buck converter. The converter regulates the voltage and maintains a stable output, typically 5 V, suitable for USB-powered devices. This process improves the reliability and efficiency of the system.

6.5 Overall System Flow

Human Effort
↓
Pedal Rotation
↓
Chain and Sprocket Mechanism
↓
100 RPM DC Geared Motor
↓
Electrical Energy Generation
↓
LM2596 Buck Converter
↓
USB Charging Module
↓
Electronic Devices

CHAPTER 7
EXPERIMENTAL SETUP AND IMPLEMENTATION

7.1 Introduction

Experimental setup and implementation are essential for validating the performance of the proposed Pedal Powered Electricity Generator. This chapter describes the fabrication, assembly, testing, and evaluation of the developed prototype. The experimental setup was designed to analyze the voltage generation capability, charging performance, and operational reliability of the system under various pedaling conditions.

The implementation phase also helps in identifying practical challenges associated with power transmission, voltage regulation, and component integration. The obtained results provide valuable information regarding the feasibility of utilizing human effort as a renewable source of electrical energy.

7.2 Prototype Description

The prototype developed for this project consists of a pedal assembly mounted on a rigid mild steel frame. A chain and sprocket mechanism connects the pedal shaft to a 100 RPM DC geared motor, which acts as the electricity generation unit. The generated voltage is regulated using an LM2596 buck converter and delivered through a USB charging module.

The overall dimensions of the prototype were selected to ensure portability and ease of operation. The system was designed to withstand continuous pedaling and provide a stable electrical output suitable for charging low-power electronic devices.

7.3 Mechanical Assembly

The mechanical assembly was carried out systematically to ensure proper alignment and efficient power transmission. The pedal mechanism was securely mounted to the frame using bolts and support brackets. Special attention was given to maintaining proper chain tension and sprocket alignment to reduce mechanical losses.

The DC geared motor was firmly attached to the frame to prevent vibration during operation. All moving parts were inspected periodically to ensure smooth rotation and reliable performance. Proper lubrication of the chain mechanism was also carried out to improve operational efficiency and increase component life.

7.4 Hardware Setup

The hardware setup comprises all electrical and electronic components required for power generation and voltage regulation. The DC geared motor generates electrical power, while the LM2596 module regulates the voltage to a constant value. The USB charging module provides a standard charging interface for electronic devices.

Additionally, an LED indicator was incorporated to provide visual confirmation of power generation. Connecting wires with appropriate insulation were used to ensure safe electrical operation. The complete hardware arrangement was assembled and tested before conducting performance evaluations.

7.5 Implementation Procedure

The implementation of the Pedal Powered Electricity Generator was performed in multiple stages. Initially, the mechanical frame was fabricated and the pedal mechanism was installed. Subsequently, the chain and sprocket arrangement was assembled, followed by the installation of the DC geared motor.

After completing the mechanical setup, the electrical components were integrated according to the circuit design. The LM2596 buck converter was calibrated to

provide a regulated output voltage of 5 V. Finally, the USB charging module was connected, and the entire system was tested under different operating conditions to verify its functionality.

7.6 Calibration and Testing

Calibration was performed to ensure accurate voltage regulation and reliable system operation. The output of the LM2596 buck converter was adjusted using a digital multimeter until a stable 5 V output was obtained. Multiple test runs were conducted to confirm the consistency of the generated voltage.

The testing procedure included no-load tests, load tests, and continuous operation tests. During experimentation, voltage readings were recorded at different pedaling speeds, and the charging performance of USB-powered devices was evaluated. The test results demonstrated that the system was capable of delivering stable and reliable power under normal operating conditions.

7.7 Experimental Procedure

The experimental procedure adopted for this project involved operating the pedal mechanism at different speeds and recording the corresponding electrical output. Initially, the prototype was inspected for proper mechanical and electrical connections. The user then pedaled the system continuously while voltage measurements were recorded using a multimeter.

After voltage generation was confirmed, the USB charging module was connected to various electronic devices to evaluate charging capability. The experiments were repeated several times to minimize errors and improve the accuracy of the collected data. The observations obtained during testing were used for performance analysis and validation of the proposed system.

7.8 Data Collection

Data collection was carried out by measuring the generated voltage at various pedaling speeds under both no-load and load conditions. The collected data were tabulated and analyzed to study the relationship between pedaling speed and electrical output.

The experimental observations revealed that the generated voltage increased with an increase in pedaling speed. Furthermore, the system exhibited stable performance when connected to low-power electronic devices through the USB charging module. The collected data were subsequently used to prepare graphs, evaluate system efficiency, and analyze the overall performance of the Pedal Powered Electricity Generator.

The successful completion of the experimental setup and implementation phase confirms the practicality and effectiveness of the proposed system. The obtained results validate the capability of converting human mechanical energy into usable electrical energy and demonstrate the potential of pedal-powered systems as sustainable and portable energy solutions.

CHAPTER 8
COST ESTIMATION

Cost estimation is an important aspect of any engineering project as it determines the economic feasibility and practical implementation of the proposed system. The Pedal Powered Electricity Generator has been designed using readily available and cost-effective components to ensure affordability without compromising performance. The use of simple mechanical and electrical components significantly reduces the overall manufacturing cost, making the system suitable for educational institutions, rural communities, and emergency applications.

The total cost of the project includes expenses associated with mechanical components, electrical components, fabrication, and assembly. The economic analysis indicates that the proposed system is considerably less expensive than conventional portable generators and does not incur recurring fuel costs. This chapter presents the cost estimation methodology, component-wise cost analysis, and economic feasibility of the developed system.

8.1 Introduction

Cost estimation plays a vital role in evaluating the financial viability of engineering projects. It provides information regarding the initial investment required for fabrication and implementation. The Pedal Powered Electricity Generator has been developed with the objective of achieving maximum performance at minimum cost.

The selection of economical and commercially available components contributes to the affordability and accessibility of the proposed system. Additionally, the absence of fuel consumption and low maintenance requirements further improve the economic benefits of the project.

8.2 Cost Estimation Methodology

The cost estimation process involves identifying all components used in the system and determining their approximate market prices. The total project cost was calculated by considering:

- Mechanical components.
- Electrical and electronic components.
- Fabrication charges.
- Assembly and installation costs.
- Miscellaneous expenses.

The component costs were estimated based on local market availability and may vary depending on supplier and location. However, the presented values provide a reasonable approximation of the overall project expenditure.

8.3 Component-Wise Cost Breakdown

S. No	Component Name	Quantity	Approximate Cost (₹)
1	Bicycle Pedal Assembly	1	800
2	100 RPM DC Geared Motor	1	650
3	LM2596 Buck Converter	1	120
4	USB Charging Module	1	80
5	Wooden Frame	1	700
6	Connecting Wires	-	100
7	Nuts, Bolts, and Fasteners	-	120

8.4 Economic Feasibility

The economic feasibility analysis indicates that the proposed system is cost-effective when compared to conventional portable generators and battery-based backup systems. Since the system utilizes human effort as its primary energy source, there are no recurring fuel expenses associated with its operation.

Furthermore, the maintenance requirements of the system are minimal and primarily involve periodic inspection of the chain mechanism, electrical connections, and supporting frame. The affordability and durability of the components contribute to the long operational life of the system.

- Low initial investment.
- Zero fuel cost.
- Minimal maintenance expenses.
- Long service life.
- Ease of component replacement.
- High portability and usability.

8.5 Comparison with Conventional Generators

The Pedal Powered Electricity Generator offers several advantages over conventional generators in terms of cost, environmental impact, and ease of operation. Unlike conventional generators, it does not require fuel for electricity generation and operates without producing harmful emissions. The system is economical, portable, and requires minimal maintenance, making it suitable for small-scale and emergency applications.

In addition, the Pedal Powered Electricity Generator operates silently and utilizes renewable human energy as its power source. Although its power output is lower than that of conventional generators, it provides a sustainable and environmentally friendly solution for charging low-voltage electronic devices.

CHAPTER 9
RESULTS AND ANALYSIS

9.1 Introduction

Results and analysis are essential for evaluating the effectiveness of the developed system. The Pedal Powered Electricity Generator was tested under different operating conditions to assess its electrical output and charging performance. The obtained data were analyzed to determine the relationship between pedaling speed and generated voltage.

The observations collected during experimentation provide valuable insights into the performance, efficiency, and practical applicability of the proposed system.

9.2 Voltage Output Analysis

The voltage output of the Pedal Powered Electricity Generator was analyzed by operating the system at different pedaling speeds. It was observed that the generated voltage increased with an increase in rotational speed of the DC geared motor. During experimentation, the system produced approximately 8 V at 40 RPM and reached nearly 30 V at 120 RPM. These results indicate that the generated voltage is directly proportional to the pedaling speed and confirm the effective operation of the generator based on the principle of electromagnetic induction.

9.3 Current Output Analysis

The current output of the system was measured under different operating conditions while charging low-power electronic devices. The results showed that the output current increased as the pedaling speed increased. At lower speeds, the generated current was sufficient for basic loads, while higher pedaling speeds provided improved charging performance. The observed increase in current

output demonstrates the capability of the Pedal Powered Electricity Generator to supply adequate electrical power for portable electronic applications.

9.4 Charging Performance

The charging performance of the Pedal Powered Electricity Generator was evaluated by connecting USB-powered devices such as mobile phones and LED lamps. The regulated 5 V output provided by the LM2596 buck converter ensured stable charging throughout the test period.

The system was capable of charging small electronic devices effectively when operated at moderate pedaling speeds. The experimental observations confirm that the proposed system is suitable for emergency charging applications and portable power generation.

9.5 Efficiency Evaluation

The efficiency of the system was evaluated by comparing the generated electrical output with the estimated mechanical input provided by the user. Factors such as transmission losses, voltage regulation losses, and user fatigue were considered during the analysis.

The overall efficiency of the Pedal Powered Electricity Generator was found to be satisfactory for small-scale electricity generation applications. The use of efficient transmission components and a high-efficiency buck converter contributed significantly to the overall performance of the system.

9.6 Graphical Representation of Results

The experimental results were graphically represented to analyze the relationship between pedaling speed and electrical output. The graphs clearly showed that both voltage and current output increased with an increase in pedaling speed. The graphical analysis provides a better understanding of the system's performance

and validates the effectiveness of the Pedal Powered Electricity Generator for low-power electricity generation applications.

9.7 Limitations

Although the Pedal Powered Electricity Generator demonstrated satisfactory performance, certain limitations were identified during testing. The generated power output is dependent on the user's pedaling speed and physical endurance. Continuous operation may lead to user fatigue, affecting the consistency of electrical output.

Additionally, the system is primarily suitable for low-power applications and cannot support high-power electrical loads. Despite these limitations, the proposed system remains a practical and sustainable solution for portable and emergency electricity generation.

In conclusion, the experimental results confirm that the Pedal Powered Electricity Generator successfully converts human mechanical energy into usable electrical energy. The system provides a reliable, environmentally friendly, and cost-effective means of generating electricity for low-voltage electronic devices, thereby demonstrating the potential of human-powered energy technologies.

CHAPTER 10
ADVANTAGES AND APPLICATIONS

The Pedal Powered Electricity Generator offers numerous advantages due to its simplicity, portability, and environmentally friendly operation. Unlike conventional generators, the proposed system does not require fossil fuels and produces no harmful emissions during electricity generation. The utilization of human effort as a renewable energy source makes the system economical and suitable for various applications.

The generated electrical power can be used for charging low-voltage electronic devices, making the system particularly useful in remote locations and emergency situations. The practical implementation of the project demonstrates the feasibility of utilizing human-powered energy for sustainable electricity generation.

10.1 Introduction

The development of renewable energy technologies has increased the demand for alternative power generation systems. The Pedal Powered Electricity Generator provides a sustainable and cost-effective solution for generating electricity through human effort. Its compact design, ease of operation, and low maintenance requirements make it suitable for a wide range of applications.

This chapter discusses the major advantages of the proposed system and highlights its potential applications in various sectors.

10.2 Advantages

The Pedal Powered Electricity Generator offers several advantages over conventional power generation systems. It is an eco-friendly system that produces electricity without generating pollutants or greenhouse gas emissions. The system

is portable and can be easily transported to different locations based on user requirements.

Furthermore, the proposed system is economical due to its low manufacturing and maintenance costs. Since it does not depend on grid electricity or fuel, it provides a reliable source of power during emergencies. The use of renewable human energy also promotes energy conservation and environmental sustainability. Additionally, the system is simple to operate, requires minimal maintenance, and serves as an effective educational tool for demonstrating renewable energy concepts.

10.3 Applications

The Pedal Powered Electricity Generator has a wide range of practical applications. It can be used as an emergency power source during electricity outages and in disaster relief operations where access to conventional power is limited. The system is also suitable for rural areas and remote locations that lack reliable electrical infrastructure.

In addition, the proposed system can be utilized in educational institutions for demonstrating principles of energy conversion and renewable energy technologies. Fitness centers can integrate the system to promote physical activity while generating electricity. The generator is also useful for camping, outdoor activities, and charging portable electronic devices such as mobile phones, LED lamps, and rechargeable batteries.

The versatility, affordability, and sustainability of the Pedal Powered Electricity Generator make it a promising solution for small-scale electricity generation and renewable energy awareness initiatives.

11. APPENDICES

11.1 FIELD VISIT



FIGURE 11.1



FIGURE 11.2



FIGURE 11.3

11.2 PROTOTYPE

FIGURE 11.4

FIGURE 11.5

FIGURE 11.6

12 .CONCLUSION

The Pedal Powered Electricity Generator project was undertaken with the objective of developing a sustainable, economical, and portable electricity generation system capable of converting human mechanical energy into usable electrical energy. The successful design and implementation of the system demonstrate the practical application of renewable energy technologies and fundamental engineering principles, including energy conversion, electromagnetic induction, power transmission, and voltage regulation. The project effectively utilizes a pedal mechanism coupled with a 100 RPM DC geared motor operating as a generator to produce electrical power. The generated voltage is regulated using an LM2596 DC-DC buck converter, ensuring a stable output suitable for charging low-voltage electronic devices through a USB charging module.The experimental analysis confirmed that the generated voltage increases proportionally with pedaling speed, thereby validating the effectiveness of the proposed design. The system was able to provide sufficient electrical power for charging mobile phones, LED lamps, rechargeable batteries, and other USB-powered devices. Furthermore, the integration of readily available mechanical and electrical components resulted in a cost-effective and easy-to-maintain system. The portability and grid-independent nature of the Pedal Powered Electricity Generator make it particularly suitable for emergency situations, rural electrification, disaster relief operations, educational institutions, fitness centers, and outdoor activities.In conclusion, the Pedal Powered Electricity Generator serves as an efficient, reliable, and sustainable solution for portable electricity generation. The successful completion of this project demonstrates the feasibility of human-powered energy systems and highlights their potential role in addressing future energy challenges. The project not only fulfills its intended objectives but also provides a foundation for further research and development in the field of renewable and human-powered energy technologies.

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